Causal Effects of PDMP Opioid Check Mandates on Illicit Opioid Mortality

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1 Introduction and Context

1.1 Motivation

The main topic for investigation in this project concerns the state-level opioid prescribing monitoring laws, specifically Prescription Drug Monitoring Program (PDMP) opioid check mandates in the U.S., which were introduced as a policy response to the opioid epidemic. The opioid epidemic was the rapid increase of opioid misuse beginning in the mid to late 1990s that has led to the deaths of hundreds of thousands of people. These monitoring-based policies had their groundwork earlier, though they reached a turning point around the mid-2010s, after which many states started more rigorously implementing and strengthening PDMP mandates (though this varied state by state).

However, although these policies were well-intentioned, there is speculation that increasing monitoring of prescription opioids may have led some individuals to turn to other illicit drugs as a result. But it is difficult to determine whether these policies directly caused individuals to seek out alternative illicit drugs, since there were also rapid changes in illicit drug supply during this time (including heroin, fentanyl, and other substances). This creates a confounding timeline problem, as it is unclear whether individuals switched to alternatives because of the policy, or whether deaths related to illicit drugs were rising independently due to other factors.

1.2 Causal Question

This motivates the analysis for the following research question: what is the causal effect of implementing a PDMP opioid check mandate on illicit opioid mortality at the state-year level? In particular, the analysis examines whether increased monitoring of prescription opioids through PDMP mandates is associated with changes in illicit opioid mortality, potentially reflecting substitution toward illicit drugs.

2 Conceptual Framework and DAG

Based on this causal question, we can create a DAG as follows:

2.1 DAG

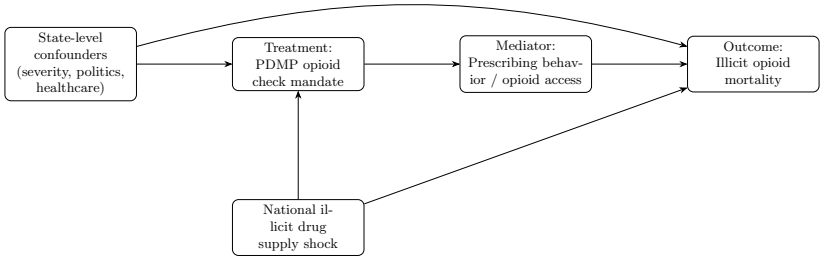


Figure 1: This is a causal DAG for the relationship between PDMP opioid check mandates and illicit opioid mortality.

2.2 DAG Explanation

From the DAG, the PDMP policy is assumed to affect illicit opioid mortality indirectly through changes in prescribing behavior and opioid access. The state-level confounders, such as opioid severity, healthcare conditions, and political factors, may influence both the likelihood of policy adoption and mortality outcomes. Additionally, the national illicit drug supply shocks may affect both treatment assignment and illicit opioid mortality.

These relationships induce back-door paths between treatment and outcome. Therefore, without the additional assumptions, the observed association between PDMP adoption and illicit opioid mortality may reflect both causal effects and confounding factors, potentially complicating identification.

3 Data Explanation

The analysis for this project consists of joining data from two sources: mortality data from the CDC and PDMP policy data for constructing the treatment variable. The first dataset contains opioid mortality data from the CDC, which reports state-level opioid-related death rates over time. In particular, the data distinguishes between prescription opioid mortality and illicit opioid mortality, allowing for separate analysis of these outcomes. Additionally, the second dataset contains information on state-level Prescription Drug Monitoring Program (PDMP) policies, including the timing of implementation of opioid check mandates requiring healthcare professionals to consult the PDMP before issuing prescriptions.

These two datasets are then merged to form a panel of U.S. states observed annually over time. To be more specific: The unit of observation is at the level

of state-year, indexed by state i and year t. Then let Y_{it} denote the illicit opioid mortality rate and P_{it} denote the prescription opioid mortality rate in state i and year t. The primary outcome variable is illicit opioid mortality, measured as the crude death rate per state per year.

The key treatment variable is an indicator for whether a PDMP opioid check mandate is in effect in a given state-year. Let G_i denote the first year in which state i implements the policy. The treatment indicator is defined as $D_{it} = \mathbf{1}(t \geq G_i)$, which equals 1 if the policy is active in state i at time t, and 0 otherwise.

3.1 Discussion of the Missing Values

A notable feature of the data is the presence of missing values in the mortality variables, particularly for illicit opioid deaths in earlier years. One thing to observe from inspecting the data is that the missing data is not uniformly distributed across states and time, as earlier years and smaller states exhibit higher rates of suppression in illicit opioid mortality. This suggests that missingness is not random and may be related to underlying outcome levels.

To mitigate this issue, the project focuses on the period after 2010, where data coverage is more comparable across states. These missing observations arise from suppression or unreliability of reported counts in the CDC data. As a result, the analysis restricts attention to the time period after 2010, where data coverage is more consistent across states. This implicitly assumes that remaining missingness does not differentially bias treated and control groups.

The final analysis sample consists of a state-year panel with multiple observations per state. This allows for the use of fixed-effects models to control for time-invariant state characteristics and common time shocks.

4 Empirical Strategy

4.1 Baseline Difference-in-Differences Specification

The staggered adoption of PDMP mandates across states provides temporal variation that can be studied using a difference-in-differences (DiD) framework. Causal interpretation, however, requires that in the absence of treatment the relevant comparison groups would have followed similar outcome trends and that policy timing is not driven by unobserved time-varying factors that also affect opioid mortality. To examine the effect of PDMP opioid check mandates on illicit opioid mortality, I estimate a two-way fixed-effects DiD specification using panel data across states and time.

To start, we can let Y_{it} refer to the illicit opioid mortality rate in state i at time t, and also let $D_{it} = \mathbf{1}(t \geq G_i)$ be an indicator equal to 1 if the PDMP mandate is in effect in state i at particular time t, where G_i denotes the year of policy adoption.

To be more specific, the baseline estimating equation is:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$$

where:

- α_i are defined as the state fixed effects, which capture the time-invariant differences across states,
- λ_t are defined as the year fixed effects, which capture the common shocks across all states,
- β is the coefficient of interest in the baseline two-way fixed-effects specification,
- ε_{it} is an error term.

State fixed effects remove time-invariant differences across states, while year fixed effects absorb shocks common to all states in a given year. These fixed effects do not, by themselves, guarantee causal identification. The causal interpretation of β instead depends on the parallel trends assumption and on the absence of differential time-varying confounding that is related to both policy adoption and opioid mortality. In addition, because treatment is adopted at different times across states, a conventional two-way fixed-effects coefficient need not correspond to a simple average treatment effect when treatment effects vary across cohorts or over time.

Additionally, one can note that the standard errors are clustered at the state level to account for serial correlation within states over time.

4.2 Identification Assumption

The key identification assumption for the DiD estimator is the parallel trends assumption, which states that in the absence of treatment, the treated and control states would have experienced similar trends in illicit opioid mortality.

To formalize the causal question, let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes with and without treatment, respectively. The substantive estimand of interest is the effect of the policy on treated states. The coefficient β from the baseline two-way fixed-effects regression is an estimator whose causal interpretation requires the identifying assumptions below; under staggered adoption and heterogeneous treatment effects, it need not map cleanly to a single ATE or ATT.

Since $Y_{it}(0)$ is not observed for treated states after treatment, the identification relies on the assumption that untreated potential outcomes evolve similarly across treated and untreated states.

This can be expressed as:

$$E[Y_{it}(0) - Y_{is}(0) \mid G_i = g] = E[Y_{jt}(0) - Y_{js}(0) \mid G_j = \infty]$$

for all t, s, where i indexes treated states, and j indexes untreated states.

Under this assumption, differences that emerge after treatment can be interpreted as policy effects rather than as continuations of pre-existing differential trends. In this setting, the assumption is most credible when policy adoption is not responding to evolving, unobserved determinants of opioid mortality that differ across states over time.

4.3 Event Study Specifications

To examine the dynamics around policy adoption and assess the plausibility of the parallel trends assumption, the analysis uses both a conventional two-way fixed-effects (TWFE) event study and a staggered-adoption event study.

The conventional TWFE event-study specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}(t - G_i = k) + \varepsilon_{it},$$

where k indexes time relative to treatment and the period $k = -1$ is omitted as the reference category. The pre-treatment coefficients ($k < 0$) provide a diagnostic for whether treated states were already on different trajectories before adoption, while the post-treatment coefficients ($k > 0$) describe how estimated differences evolve after adoption.

Because treatment is adopted at different times across states, conventional TWFE event-study coefficients can be difficult to interpret when treatment effects vary across cohorts or over time. To address this concern, the analysis also estimates a Sun & Abraham staggered-adoption event study using cohort-by-relative-time comparisons. This specification is designed to avoid problematic comparisons between already-treated and newly treated units and allows treatment effects to vary across adoption cohorts and relative time.

The Sun & Abraham estimates are therefore treated as the preferred dynamic specification, while the conventional TWFE event study is retained as a diagnostic comparison.

4.4 Potential Threats to Identification

After analyzing the available data, a potential major concern in this configuration is that policy adoption may actually be endogenous instead of exogenous. States experiencing rising opioid misuse or mortality may be more likely to implement PDMP mandates earlier, violating the parallel trends assumption.

This pattern would indicate that policy timing is related to evolving determinants of opioid mortality, undermining the identifying assumptions required for a causal DiD interpretation.

Additionally, the timing of PDMP adoption overlaps with broader national changes in illicit opioid supply, including the rapid increase in fentanyl availability. These concurrent shocks may differentially affect states and confound the estimated treatment effects.

The presence of non-random missing data in earlier years further complicates identification, as missingness is correlated with state size and outcome levels.

4.5 Robustness and Sensitivity Analyses

To assess the robustness of the results, the analysis considers several complementary specifications and diagnostics:

- **Sun & Abraham staggered-adoption event study:** allowing treatment effects to vary across adoption cohorts and relative time,
- **Restricted sample analysis:** limiting the sample to early-adopting states to assess sensitivity to treatment-timing composition,
- **State-specific trend model:** allowing states to follow separate linear time trends,
- **Pre-treatment diagnostics:** examining event-study coefficients before policy adoption to assess the plausibility of parallel trends.

These analyses help evaluate the credibility of the identification strategy and the sensitivity of the findings to modeling choices.

5 Descriptive Evidence

We can start by observing the descriptive trends in opioid mortality across states over time. The figure 2 presents the average prescription opioid mortality P_{it} and illicit opioid mortality Y_{it} over the sample period.

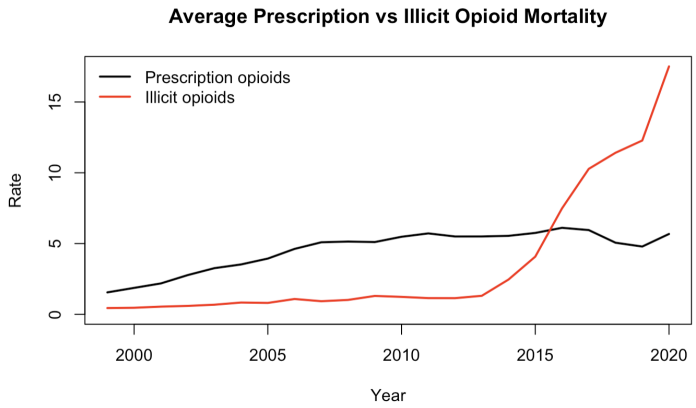


Figure 2: Average prescription and illicit opioid mortality over time. Illicit mortality rises sharply after 2013, while prescription mortality remains relatively stable or declines.

The data reveals that prescription opioid mortality remains relatively stable over time (however with some evidence of decline in later years). In contrast to this, the illicit opioid mortality shows a sharp increase, after around 2013. Interestingly, this period seems to coincide with a major change in the illicit drug supply, including the emergence of synthetic opioids such as fentanyl.

To further examine the differences across the treatment groups, we can also compare the trends in illicit opioid mortality between both the treated and

untreated states. To be clear, define D_{it} to the treatment status, and let $D_i = 1$ refer to the states that ever adopt the policy.

The treated states tend to exhibit higher levels of illicit opioid mortality even prior to treatment. Differences in baseline levels do not by themselves violate the parallel trends assumption, but they indicate that treated and untreated states are systematically different. More importantly, the descriptive trends also appear to differ before policy implementation, motivating a formal examination of pre-treatment dynamics in the event-study analysis.

Thus, the descriptive evidence raises concerns about comparability across groups, while the presence of differential pre-treatment trends would directly threaten the parallel trends assumption required for causal identification.

6 Main Results

Now, the results of the project can be discussed.

6.1 Baseline Difference-in-Differences Estimates

First, an estimate of the baseline difference-in-differences model can be constructed in the following way:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \varepsilon_{it}$$

where β is the coefficient on the treatment indicator in the baseline two-way fixed-effects specification. A causal interpretation of this coefficient depends on the identifying assumptions discussed above and should be treated cautiously under staggered adoption with potentially heterogeneous treatment effects.

It can also be seen that the estimated coefficient on D_{it} is positive, however, not statistically significant, when the standard errors are clustered at the same state level.

Table 1: Baseline DiD Estimates of PDMP Effect on Illicit Mortality

	Estimate	Std. Error
Treatment (D_{it})	1.30068	1.57863

The estimate is positive but statistically imprecise with state-clustered standard errors. Given the identification concerns discussed below, this coefficient should not be interpreted as credible evidence either for or against a causal effect of PDMP mandates on illicit opioid mortality.

6.2 Event Study Analysis

To examine the dynamics of treatment effects and test the parallel trends assumption, we can estimate:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}(t - G_i = k) + \varepsilon_{it}$$

Figure 3 presents the estimated coefficients β_k.

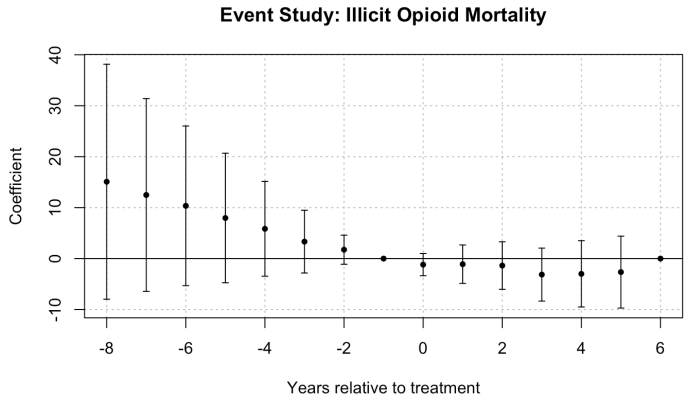


Figure 3: This is the event study estimates of treatment effects. Pre-treatment coefficients deviate from zero, which indicates a violation of the parallel trends assumption.

Several pre-treatment coefficients ($k < 0$) are statistically different from zero, providing evidence that treated states were already on different trajectories compared with the relevant comparison states prior to policy implementation.

Formally, this implies:

$$E[Y_{it}(0) - Y_{is}(0) \mid D_i = 1] \neq E[Y_{jt}(0) - Y_{js}(0) \mid D_j = 0]$$

which is inconsistent with the parallel trends assumption required for the baseline DiD interpretation.

The post-treatment coefficients also do not display a consistent or stable pattern. Because the event study is estimated using a conventional two-way fixed-effects specification under staggered adoption, these coefficients are best treated as diagnostic rather than as clean cohort-specific causal effects.

6.3 Sun & Abraham Staggered-Adoption Event Study

Because PDMP mandates were adopted at different times across states, the analysis also estimates a Sun & Abraham event-study specification that allows treatment effects to vary across adoption cohorts and relative time. This approach avoids relying on comparisons between already-treated and newly treated states that can complicate conventional TWFE event-study estimates.

The staggered-adoption estimates are used as the preferred dynamic specification. Pre-treatment estimates remain important for assessing whether the untreated potential outcomes of treated and comparison states appear to follow

similar trajectories before adoption. If meaningful pre-treatment differences remain under this specification, the main identification concern is not resolved by changing estimators alone.

The Sun & Abraham analysis therefore complements, rather than replaces, the identification diagnosis from the conventional event study: a more appropriate staggered-adoption estimator can address treatment-timing and heterogeneity concerns, but it cannot recover a credible causal effect if policy adoption is itself related to differential pre-treatment outcome trends.

6.4 State-Specific Trend Model

As an additional robustness check, we can also estimate the following:

$$Y_{it} = \alpha_i + \lambda_t + \delta_i t + \beta D_{it} + \varepsilon_{it}$$

where $\delta_i t$ captures the particular state-specific linear trends.

Table 2: DiD Estimates with State-Specific Trends

	Estimate	Std. Error
Treatment (D_{it})	-0.494453	0.382731

Here, the estimated effect becomes smaller and changes sign, suggesting that baseline results may reflect differential trends rather than a causal effect.

6.5 Sensitivity to Sample Restrictions

We can restrict the sample to early-adopting states to attempt to reduce the observed variation in treatment timing.

Table 3: DiD Estimates: Early Adopters Only

	Estimate	Std. Error
Treatment (D_{it})	4.63105	2.65881

Here, the estimated effect is larger but imprecise, indicating sensitivity to sample selection and further weakening causal interpretation.

The estimated effects seem to vary across specifications, indicating sensitivity to sample selection.

6.6 Summary

Taken together, the results indicate that estimated policy effects are sensitive to specification and sample choices. The Sun & Abraham staggered-adoption analysis improves on the conventional TWFE event-study design by accommodating heterogeneous treatment timing, but it does not remove the broader

identification concerns created by differential pre-treatment trends and potentially endogenous policy adoption.

Combined with the previously mentioned evidence of pre-treatment differences, these findings raise concerns about the credibility of a causal interpretation.

7 Diagnosis

The empirical results suggest that the identifying assumptions required for a causal interpretation of the DiD estimates are unlikely to hold in this setting.

The conventional event study reveals non-zero pre-treatment coefficients, indicating that treated and comparison states exhibit differential trends in illicit opioid mortality even prior to policy adoption. The Sun & Abraham staggered-adoption specification addresses concerns about heterogeneous treatment timing, but the central identification problem remains if pre-treatment trajectories differ systematically across groups. This is inconsistent with the parallel trends condition required for causal interpretation.

Additionally, treated states tend to have higher levels of illicit opioid mortality before treatment. Baseline level differences alone do not violate parallel trends, but together with the evidence of differential pre-treatment trajectories they are consistent with selection into treatment: states experiencing more severe or rapidly worsening opioid conditions may have been more likely to adopt PDMP mandates.

Finally, we can note that the timing of policy adoption is highly clustered in the mid-2010s, which coincides with a period of rapidly increasing illicit opioid deaths driven in part by the emergence of fentanyl. This interesting overlap suggests that policy implementation may be a response to increasingly worse outcomes rather than an outside, exogenous intervention.

Taken together, these findings suggest that policy timing may be endogenous: unobserved time-varying factors may influence both adoption and opioid mortality. As a result, the baseline DiD estimates cannot be interpreted as causal effects with confidence.

Rather than providing evidence of a true zero effect, the results instead highlight the difficulty of isolating causal effects in a setting where policy adoption is closely tied to underlying outcome trends.

8 Extension and Alternative Design

The analysis already incorporates a Sun & Abraham staggered-adoption event-study specification to address limitations of conventional TWFE estimators under heterogeneous treatment timing. Future work could compare these estimates with alternative staggered-adoption approaches such as Callaway and Sant'Anna, particularly to assess the sensitivity of cohort-specific and aggregate treatment-effect estimates.

A second direction would be to seek plausibly exogenous variation in exposure to changes in the illicit opioid supply. The national rise in fentanyl availability around 2013 is an important concurrent shock, but its timing alone does not provide sufficient identifying variation because it affects the entire country and state-level exposure may itself be endogenous. A stronger design would require an external source of differential exposure across states that is plausibly unrelated to their underlying mortality trends.

Another approach would be to examine substitution patterns between prescription and illicit opioids. Joint analysis of prescription-opioid and illicit-opioid mortality could provide additional evidence on whether reductions in prescription-related harm are accompanied by increases in illicit-drug mortality, while still requiring careful attention to the same identification concerns.

9 Conclusion

This project investigated the relationship between PDMP opioid check mandates and illicit opioid mortality using a difference-in-differences framework applied to state-year panel data.

The results do not provide credible evidence of a causal effect of PDMP mandates on illicit opioid mortality. While the baseline estimate is positive, it is statistically imprecise with state-clustered standard errors and sensitive to model specification. Moreover, the event-study analysis provides evidence of differential pre-treatment trends, and the broader descriptive patterns raise concerns about endogenous policy adoption.

These findings are consistent with the possibility that states implemented PDMP mandates in response to worsening opioid conditions rather than independently of the underlying trends. As a result, the identifying assumptions required for causal interpretation are unlikely to hold, and the estimated coefficients should not be interpreted as causal effects.

More broadly, the analysis illustrates the challenges of causal policy evaluation in observational settings where treatment is not randomly assigned. In the context of the opioid epidemic, overlapping policy interventions, heterogeneous adoption timing, and rapid changes in the illicit drug supply make causal relationships difficult to isolate.

Future work should focus on alternative research designs that exploit more credible sources of identifying variation, comparisons with additional staggered-adoption estimators such as Callaway and Sant'Anna, and improved data that better capture underlying mechanisms. Despite these limitations, the analysis provides useful evidence about why simple policy comparisons can be misleading when adoption is related to evolving outcome trends.

9.1 Code

The code can be found here:

`https://github.com/SriP15/Healthcare-Policy-Effects-on-Opioid-Mortality`